



# ACADMATE

# AI-Powered Personalized Academic Advisement System

SPRINT 0

**From early risk detection to explainable,  
personalized academic support.**

Student Data → Risk Prediction → Explainability →  
Recommendations → Human Support

# Agenda

Sprint 0 presentation roadmap

Sprint 0

1

**Team Members, Roles & Responsibilities**

2

**Problem Statement**

3

**Project Description & Solution Workflow**

4

**User Personas**

5

**Minimum Viable Product (MVP)**

6

**System Architecture & Technologies**

7

**Project Schedule / Sprint Cadence**

8

**Team Working Agreement & Definition of Done**

9

**Sprint 0 Retrospective**

# Team Members & Roles

Scrum responsibilities with clear technical ownership

Sprint 0

T

Tafadzwa

Product Owner & AI/ML  
Developer

P

Pranali

Scrum Master & Backend  
Developer

L

Lyno

Frontend Developer&LLM  
Engineer

L

Liangxian Li

Backend Developer

D

Datta

Database Manager & Backend  
Developer

S

Saisai

UI/UX Designer & Documentation  
Lead

C

Chintu

Frontend Developer & QA Tester

# Problem Statement

Why academic advising needs a more proactive, data-informed approach

## ! Current Challenge

Universities collect grades, attendance, assignment, course-performance and LMS-engagement data, but these signals are often fragmented across systems.

Students may discover academic problems only after performance has already declined, while advisors must manually identify who needs support and why.

## ✓ What is Needed

**A unified advising platform that moves beyond displaying academic data.**

- Identify students who may need support
- Explain the factors contributing to academic risk
- Recommend appropriate, personalized actions
- Help advisors intervene earlier
- Keep humans involved in important decisions

# Project Description

A full-stack advising platform that turns academic data into actionable support

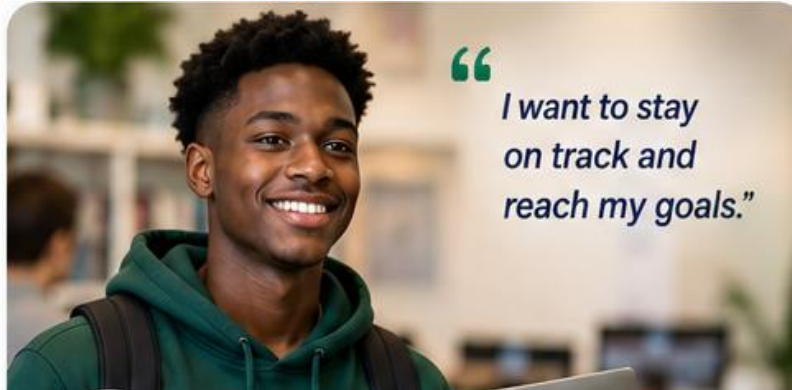


# Student Persona

Learn • Improve • Succeed

AI-Powered Personalized  
Academic Advisement System

Sprint 0



## Description

A current student who wants a clear view of academic progress, upcoming work, risk indicators and practical next steps.



## Goals

- Track grades, attendance and progress
- Understand why support is recommended
- Receive personalized actions and resources
- Ask academic questions through the AI advisor



## Challenges

- Fragmented academic information
- Late awareness of declining performance
- Generic rather than personalized support

# Academic Advisor Persona

Guide • Support • Make an Impact

AI-Powered Personalized  
Academic Advisement System

Sprint 0



## Description

An academic advisor who needs to quickly identify students requiring attention and understand the evidence behind recommendations.

## ✓ Goals

- Monitor assigned students
- Prioritize students needing intervention
- Review risk explanations and recommendations
- Record interventions and follow-up

## ! Challenges

- Large advising caseloads
- Manual review of multiple data sources
- Limited time for proactive outreach



# Administrator Persona

Manage • Enable • Ensure Success



“  
I want a safe,  
reliable system  
that improves  
student outcomes.”

## Description

A university administrator who needs to manage the platform, academic resources, access controls and system-level reporting.

## Goals

- Manage users, roles and resources
- Maintain approved advising content
- Monitor system usage and outcomes
- Support governance and auditability

## Challenges

- Data privacy and access control
- Consistency of advising information
- Responsible use of AI

# Minimum Viable Product (MVP)

The smallest complete system that demonstrates end-to-end academic support

ID

## Authentication & Roles

Student, advisor and  
administrator access

AP

## Academic Profile

Courses, grades, attendance and  
assignments

SD

## Student Dashboard

Progress, risk status and  
recommendations

AD

## Advisor Dashboard

Students, explanations and  
interventions

ML

## Risk Prediction

ML-based academic-support risk

XAI

## Explainable AI

Show factors influencing the  
result

RE

## Recommendation Engine

Map needs to actions and  
resources

IT

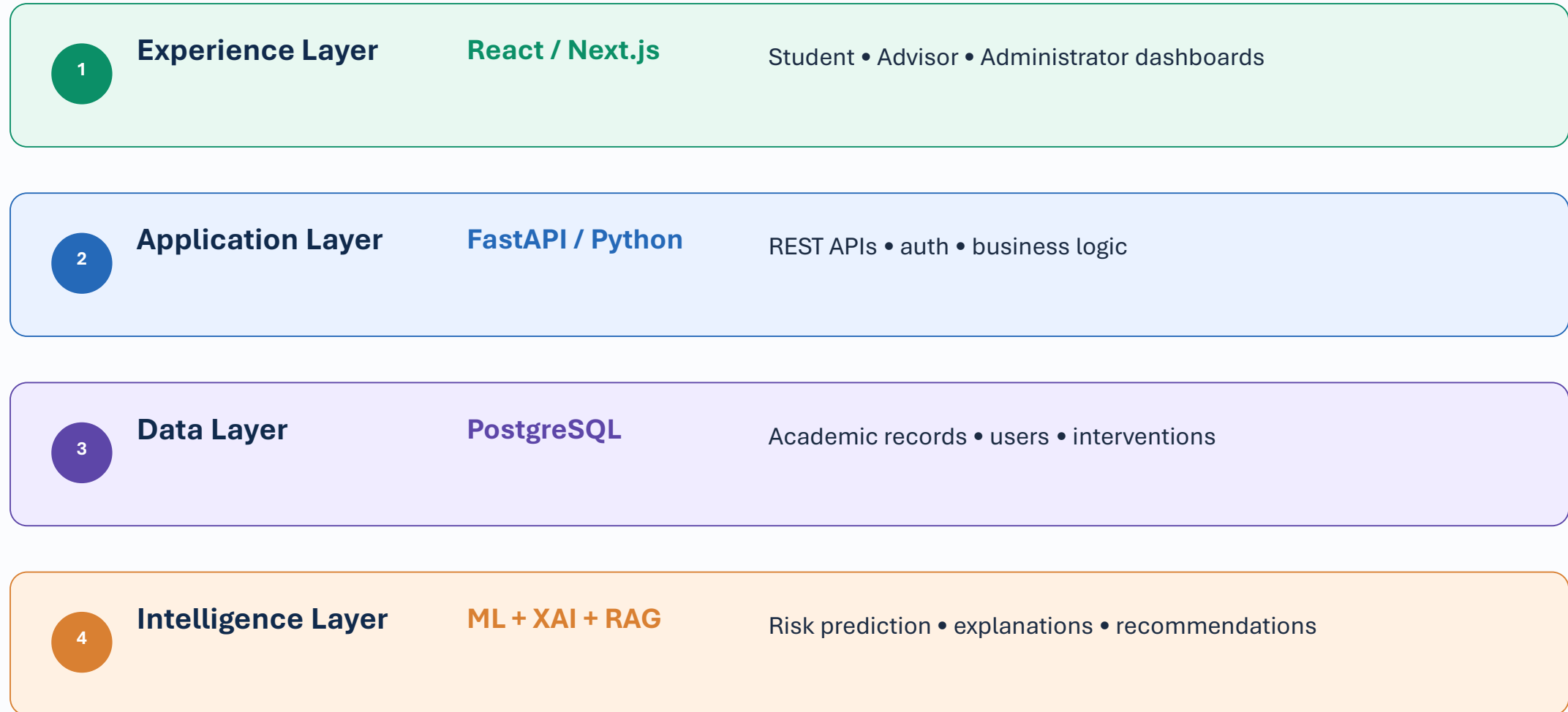
## Intervention Tracking

Review actions and monitor  
follow-up

# Proposed System Architecture

Separation between experience, application, data and intelligence layers

Sprint 0



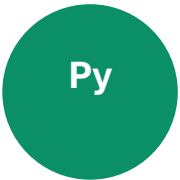
# Technologies & AI/ML

A practical stack aligned with the team’s skills and project scope



**React / Next.js**

Frontend



**Python / FastAPI**

Backend



**PostgreSQL**

Database



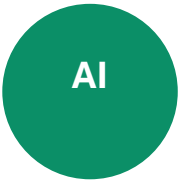
**scikit-learn**

Machine Learning



**SHAP**

Explainability



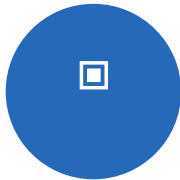
**LLM + RAG**

Conversational AI



**GitHub**

Collaboration



**Docker + CI/CD**

DevOps

# Project Schedule / Sprint Cadence

Each sprint delivers a working increment toward the final MVP

Sprint 0

0

## Sprint 0 — Discovery & Planning

Requirements • personas • architecture • backlog • working agreement

1

## Sprint 1 — Core Advisement Platform

Authentication • academic data • APIs • student/advisor dashboards

2

## Sprint 2 — Intelligence & Intervention

Risk prediction • explainability • recommendations • interventions

3

## Sprint 3 — AI, Integration & Release

Conversational AI/RAG • testing • security • deployment

# Team Working Agreement & Definition of Done

How we will collaborate and what “complete” means

## Team Principles

- Respectful, timely and transparent communication
- Clear ownership with cross-component collaboration
- Balanced workload and early escalation of blockers
- Use GitHub issues, branches and pull requests
- Attend Scrum ceremonies and give honest status updates
- Document technical decisions and important changes
- Integrate frequently rather than waiting until sprint end

## Definition of Done

- User story meets acceptance criteria
- Code runs without known blocking errors
- Appropriate tests are completed and passing
- Peer review is completed before merge
- Database, API and UI changes are integrated
- Documentation is updated where required
- Product Owner verifies the completed story
- Work is demonstrated during the sprint review

# Sprint 0 Retrospective

What we learned from discovery and what happens next

Sprint 0

## What Went Well

- Agreed on a capstone topic
- Defined the product direction
- Assigned Scrum and technical roles
- Identified three primary personas
- Established preliminary MVP and architecture

## What to Improve

- Finalize detailed user stories and acceptance criteria
- Confirm data source / simulated dataset strategy
- Finalize UI wireframes
- Agree on deployment environment
- Refine AI scope to remain achievable

## Next Actions

- Complete product backlog
- Finalize ERD and API contracts
- Create initial UI wireframes
- Set up repository and branch strategy
- Prepare Sprint 1 backlog
- Begin core platform implementation

# Sprint 1 Planning — Core Advisement Platform

Build the first functional student/advisor experience on top of the Sprint 0 foundation

1

Sprint Goal

Deliver a working core platform where students and advisors can sign in, view essential academic information, and navigate initial dashboards.

Sprint 1 Backlog

Prioritized work for the sprint

AUTH

Authentication & Roles

Student, advisor and admin sign-in with role-based access.

DATA

Academic Data

Create core academic profile structure for courses, grades and attendance.

API

Backend APIs

Implement core services that connect academic data to the user experience.

UI

UI/UX Wireframes

Design the first user flows and dashboard wireframes for key roles.

STU

Student Dashboard

Show progress, academic status and personalized support information.

ADV

Advisor Dashboard

Surface students needing attention, explanations and intervention context.

INT

Frontend–Backend Integration

Connect dashboard components with APIs and shared data models.

TEST

Basic Testing

Validate integration, usability and critical user flows before review.

Expected Sprint Outcome

Sign-in works

Academic profile ready

Dashboards visible

Core APIs connected

Key flows tested



# THANK YOU

AI-Powered Personalized Academic Advisement System